

Continual Learning and Model Editing for Foundation Models: A Survey

Foundation models are deployed in environments where knowledge, user needs, policies, and data distributions change after pretraining. Continual learning and model editing address this post-deployment adaptation problem from complementary directions: continual learning updates capabilities over streams of tasks or data, while model editing targets localized factual, behavioral, or safety-relevant changes without full retraining. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around lifelong adaptation, catastrophic forgetting, replay and regularization, parameter-efficient modules, knowledge editing, localization of internal representations, evaluation, unlearning, safety, and multimodal foundation models. The survey argues that future foundation-model maintenance will require a unified view of plasticity, locality, generalization, provenance, and governance: models must change when evidence changes, preserve unrelated capabilities, expose the scope of each update, and support auditing after deployment.

Keywords:

continual learning; model editing; foundation models; large language models; catastrophic forgetting; knowledge editing; machine unlearning; evaluation

1 Introduction

Foundation models are trained on large, heterogeneous corpora, but they are rarely static artifacts after release. Facts change, domains drift, software libraries evolve, policies are revised, private data must be removed, and users expect systems to incorporate feedback without losing previously useful behavior. Continual learning and model editing therefore study how a pretrained model can be maintained after its initial training run [26, 92, 145, 165, 177, 201, 211, 281]. The problem is especially pressing for large language models and multimodal foundation models because full retraining is expensive, slow, and often incompatible with rapid updates.

Continual learning asks a model to learn from a sequence of tasks, domains, instructions, or data streams while preserving earlier capabilities. Model editing asks for a targeted update, such as changing a fact, correcting a harmful behavior, or removing sensitive knowledge, while leaving unrelated predictions intact. The two traditions have different experimental roots, but deployment collapses the distinction:

a maintained foundation model may need many local edits, periodic continual pretraining, parameter-efficient specialization, and explicit unlearning over the same lifetime [26, 41, 92, 145, 165, 177, 201, 211].

This survey reviews the literature through the lens of foundation-model maintenance. We focus on methods that address plasticity without uncontrolled forgetting, locality without brittle memorization, and adaptation without losing calibration or safety. The central claim is that continual learning and model editing should be evaluated as update systems rather than isolated algorithms: an update has a trigger, evidence, scope, mechanism, validation protocol, audit trail, and rollback strategy.

2 Background

2.1 Foundation Models as Mutable Systems

The classical machine-learning pipeline treats training as a one-time procedure followed by deployment. Foundation models make this view unrealistic

tic. Their outputs encode factual associations, procedural skills, reasoning habits, instruction following policies, and domain-specific conventions. Once deployed, the model interacts with new data and new constraints. Updating all parameters with new data may cause catastrophic forgetting; refusing to update leaves stale or unsafe behavior in place. The maintenance problem therefore sits between pretraining, fine-tuning, retrieval, editing, and governance [26, 92, 145, 165, 177, 201, 211, 281].

2.2 Continual Learning

Continual learning is concerned with sequential adaptation. A model may encounter tasks one by one, receive a stream of documents, adapt to changing user distributions, or learn new instructions after alignment. Core questions include how to balance stability and plasticity, how to allocate capacity for new knowledge, how to rehearse old behavior without storing all old data, and how to measure transfer and forgetting. In foundation models, these questions interact with scale, instruction tuning, long-context use, and parameter-efficient adaptation [26, 41, 92, 145, 165, 177, 201, 211].

2.3 Model Editing

Model editing targets local changes to model behavior. A typical factual edit might ask a language model to replace an outdated subject-relation-object association, while preserving unrelated facts and paraphrase generalization. Editing methods include gradient updates, hypernetworks, memory-based routing, rank-one updates, locate-and-edit procedures, and external editor modules. The desired properties are reliability on the edit, locality outside the edit scope, generalization to rephrasings, and portability across contexts [4, 26, 41, 66, 94, 233, 279, 281].

2.4 Forgetting, Interference, and Locality

Both continual learning and model editing face interference. Continual updates can overwrite old skills, and local edits can unexpectedly change nearby facts or reasoning patterns. Catastrophic forgetting is usually measured at the level of tasks or datasets, whereas edit locality is measured at the level of neighboring prompts, subjects, relations, or behaviors. The shared technical issue is credit assignment: which parameters, activations, memories, or modules should change in response to a new piece of evidence [26, 41, 92, 145, 177, 201, 211, 281].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with continual learning, lifelong learning, incremental learning, continual pretraining, continual instruction tuning, catastrophic forgetting, replay, rehearsal, parameter-efficient adaptation, LoRA, adapters, model editing, knowledge editing, ROME, MEMIT, MEND, SERAC, CounterFact, ZsRE, KnowEdit, KEBench, causal tracing, machine unlearning, privacy, safety, multimodal foundation models, vision-language models, and code language models. Records were retained when their title or abstract linked language models, large language models, or foundation models to continual adaptation, editing, forgetting, knowledge updates, unlearning, evaluation, or related mechanisms. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 A Taxonomy of Foundation-Model Updates

We organize the field around six update axes. First, the *update granularity* may be a single fact, a safety rule, a domain corpus, a user preference, a new task, or a stream of tasks. Second, the *mechanism* may update all weights, a small parameter-efficient module, a memory, a router, a retrieval index, or a localized internal component. Third, the *scope* describes what should change and what should remain invariant. Fourth, *evidence* defines the source of the update: demonstrations, counterfactual facts, documents, human feedback, policy changes, or deletion requests. Fifth, *evaluation* measures reliability, locality, generalization, retention, robustness, and cost. Sixth, *governance* records provenance, privacy obligations, authorization, and rollback.

3.3 Continual Pretraining and Instruction Adaptation

Continual pretraining extends a foundation model with new corpora or domains. It can improve temporal freshness and domain coverage, but it risks overwriting rare skills or changing instruction-following behavior. Continual instruction tuning is more targeted: the model receives new tasks, preferences, or demonstrations after an aligned checkpoint. The main design choices are data mixing, learning-rate schedules, rehearsal sources, curriculum order, and

whether updates should be global or confined to adapters and other modules [26, 41, 92, 145, 165, 177, 201, 211].

3.4 Forgetting Mitigation

Forgetting mitigation methods include replay buffers, synthetic rehearsal, regularization, constraints on parameter movement, distillation from previous checkpoints, task-specific modules, and routing. Foundation models complicate these methods because old capabilities are broad and difficult to enumerate. A model can retain benchmark accuracy while losing long-tail factual knowledge, stylistic control, or safety behavior. Evaluation must therefore test both explicit old tasks and implicit invariants such as calibration, refusal behavior, and factual consistency [26, 41, 92, 145, 177, 201, 211, 281].

3.5 Parameter-Efficient and Modular Adaptation

Parameter-efficient fine-tuning reduces update cost by training a small subset of parameters or adding lightweight modules. LoRA, adapters, prompt tuning, prefix tuning, and mixture-of-expert routing can isolate new knowledge and allow multiple updates to coexist. The benefit is operational: updates become cheaper, easier to version, and easier to roll back. The challenge is composition. Multiple modules can interfere, routing can fail under distribution shift, and localized capacity may be insufficient for broad conceptual changes [26, 41, 108, 145, 165, 177, 201, 211].

3.6 Knowledge Editing

Knowledge editing methods attempt to modify specific model associations without full retraining. Some approaches learn an editor that predicts parameter updates from an edit example; others locate internal states associated with a fact and apply a constrained update. Memory-based approaches attach edited behavior to a retrieval or routing mechanism. The core trade-off is reliability versus locality: aggressive updates satisfy the edit but can damage nearby knowledge, while conservative updates preserve locality but may fail on paraphrases or multi-hop contexts [4, 26, 41, 66, 94, 233, 279, 281].

3.7 Localization and Mechanistic Evidence

Editing often depends on assumptions about where knowledge is stored. Causal tracing, activation

patching, knowledge neuron analyses, attribution methods, and representation studies try to identify the components most responsible for a fact or behavior. These methods are valuable because they turn editing from blind optimization into a more targeted operation. Yet localization is not a full explanation: the same behavior can be distributed across layers, depend on context, or reappear through correlated representations [7, 8, 18, 104, 137, 156, 165, 281].

3.8 Evaluation Protocols

Evaluation is the hardest part of foundation-model updates. For continual learning, metrics should measure forward transfer, backward transfer, forgetting, robustness, and update cost. For model editing, metrics should measure edit success, paraphrase generalization, locality, portability across contexts, sequential-edit behavior, and resistance to overfitting. Benchmarks such as factual-editing datasets are useful, but deployment also requires temporal freshness, multi-hop consistency, safety behavior, and audits over long update chains [41, 92, 145, 165, 177, 201, 211, 281].

3.9 Unlearning, Privacy, and Safety Updates

Machine unlearning asks a model to remove the influence of data, facts, copyrighted content, or private information. Safety updates ask a model to change behavior under harmful or policy-sensitive prompts. These updates differ from ordinary adaptation because the desired behavior may be a refusal, a deletion, or a bounded transformation rather than a new positive capability. Evaluation must verify that removed information is not recoverable through paraphrase, jailbreak, or composition with other context [11, 26, 92, 108, 112, 165, 233, 281].

3.10 Multimodal and Domain-Specific Foundation Models

Continual adaptation and editing are not limited to text. Vision-language models, code models, scientific models, and robotic foundation models face changing taxonomies, APIs, environments, and user goals. Multimodal updates raise additional locality questions because a visual concept, a text label, and an action affordance may be entangled. Domain models also need update records that satisfy scientific, medical, legal, or software-engineering constraints [30, 92, 108, 145, 153, 165, 201, 211].

3.11 Representative Literature Map

The following map cites the full retained corpus by theme. It makes the survey coverage explicit and separates continual-learning, editing, evaluation, unlearning, and multimodal strands.

Continual pretraining and lifelong adaptation The continual pretraining and lifelong adaptation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [3, 81, 136, 154, 214, 217, 272, 282, 292]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [1, 9, 10, 68, 79, 98, 129, 140, 243]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [269, 274].

Catastrophic forgetting and replay mechanisms The catastrophic forgetting and replay mechanisms cluster contains work that develops methods, systems, datasets, or analyses central to this survey [11, 44, 108, 122, 153, 177, 201, 202, 247]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [22, 43, 69, 123, 128, 215, 234, 248, 293]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [21, 35, 126, 131, 134, 143, 199, 200, 278]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [64, 72, 142, 159, 171, 197, 209, 220, 241]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [24, 37, 51, 59, 164, 187, 205, 235, 298]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [16, 76, 105, 163, 174, 180, 192, 194, 283]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [46, 52, 95, 102, 182, 183, 203, 212, 229].

Parameter-efficient and modular adaptation The parameter-efficient and modular adaptation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [20, 26, 45, 141, 148, 184, 237, 261, 275]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [40, 88, 96, 103, 110, 114, 127, 208, 225].

Model editing and knowledge updates The model editing and knowledge updates cluster contains work that develops methods, systems, datasets, or analyses central to this survey [4, 41, 47, 66, 77, 112, 130, 156, 281]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [7, 8, 34, 57, 93, 137, 193, 223, 276]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [17, 117, 133, 162, 166, 168, 172, 173, 228]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 56, 73, 75, 87, 124, 236, 284, 290]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [27, 28, 84, 89, 147, 160, 196, 240, 245]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [2, 29, 58, 71, 146, 150, 178, 256, 286]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [19, 39, 49, 62, 109, 115, 167, 206, 224]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [38, 42, 48, 138, 151, 222, 246, 270, 285]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [31, 63, 107, 135, 219, 239, 242, 244, 250].

Localization, attribution, and internal representations The localization, attribution, and internal representations cluster contains work that develops methods, systems, datasets, or analyses central to this survey [18].

Benchmarks, datasets, and evaluation The benchmarks, datasets, and evaluation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [92, 94, 100, 145, 165, 169, 226, 260, 279]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [23, 25, 83, 104, 118, 144, 191, 204, 294]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [14, 55, 86, 97, 185, 186, 262, 264, 267]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [65, 80, 158, 190, 207, 232, 254, 271, 273]. Additional

retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [179, 195, 213, 216, 230, 255, 263, 291, 297]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [5, 33, 67, 113, 139, 252, 288, 295, 296]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [12, 53, 70, 101, 106, 120, 121, 218, 221]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [50, 82, 125, 152, 155, 238, 257, 266, 289]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [36, 60, 78, 85, 111, 157, 189, 227, 265]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [32, 61, 119, 181, 198, 210, 231, 253, 259]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [90, 99, 170, 175, 299].

Safety, privacy, and machine unlearning The safety, privacy, and machine unlearning cluster contains work that develops methods, systems, datasets, or analyses central to this survey [54, 132, 233, 280].

Multimodal and domain-specific foundation models The multimodal and domain-specific foundation models cluster contains work that develops methods, systems, datasets, or analyses central to this survey [15, 30, 149, 161, 176, 211, 249, 277, 287]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [13, 74, 91, 116, 188, 251, 258, 268, 300].

4 Challenges and Prospects

4.1 Sequential Editing and Update Composition

Most model-editing experiments evaluate one edit or a small batch, but deployed systems accumulate many updates. Future methods need guarantees about sequential composition: later edits should not erase earlier repairs, modules should not contradict one another, and update logs should explain which mechanism controls a response.

4.2 Locality Beyond Nearby Prompts

Locality is often tested with unrelated prompts, but real models contain latent associations. A fact edit can change reasoning, persona, translation, or safety behavior in unexpected ways. Stronger evaluation should include semantic neighborhoods, multi-hop questions, adversarial paraphrases, and long-context interactions.

4.3 Plasticity Without Unbounded Drift

Continual learning requires plasticity, but foundation models also serve as stable platforms for applications. The key challenge is to adapt when evidence is trustworthy and resist drift when data are noisy, malicious, temporary, or unauthorized. This points toward update controllers that combine provenance, uncertainty, validation, and human approval.

4.4 Auditable Maintenance Pipelines

Foundation-model updates need operational infrastructure: issue tracking, evidence storage, authorization, test suites, versioned adapters, rollback, and post-update monitoring. Research on continual learning and model editing should connect more directly with these maintenance pipelines so that updates can be reproduced and audited.

4.5 Retrieval, Editing, and Training as a Unified Stack

Some knowledge should live in retrieval indexes, some in editable modules, and some in core model parameters. The future is likely a hybrid stack in which retrieval handles volatile evidence, model editing handles targeted persistent changes, and continual training handles broad distributional shifts. The open question is how to decide which layer should absorb each update.

5 Conclusion

Continual learning and model editing address a central problem for foundation models: how to change after deployment without uncontrolled forgetting or hidden side effects. This survey reviewed life-long adaptation, forgetting mitigation, parameter-efficient modules, knowledge editing, localization, evaluation, unlearning, and multimodal extensions. The main lesson is that update quality cannot be judged by immediate success alone. A good update must be reliable, local, generalizable, auditable, reversible when necessary, and compatible with future

updates. Treating foundation models as maintained systems rather than fixed artifacts will make continual learning and model editing increasingly important for safe and useful AI.

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